Data Analytics – Exercises

(Week 12)

In these exercises, you will learn:

- how to build and train a simple neural network from scratch using NumPy, including forward propagation and backpropagation
- how a neural network updates its weights step by step during training (manual walk-through of one backpropagation update)
- how to build and train a Convolutional Neural Network (CNN) with Keras/TensorFlow to recognize handwritten digits (MNIST)
- how to build and train a neural network with Keras/TensorFlow to classify images of clothing (Fashion MNIST), and interpret training vs. test accuracy

In the data analytics process model, these exercises cover part of the step "Modeling" (see slides of week 01).

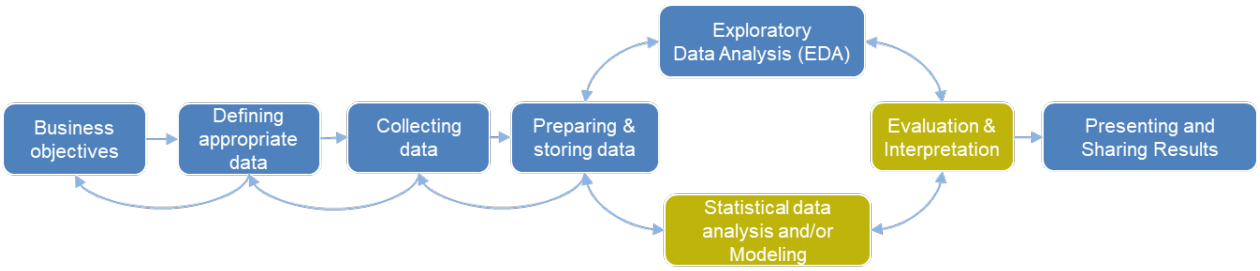


Figure 1: Data analytics process model (see slides of week 01)

Task 1

In this exercise, you will build and train a simple neural network from scratch using only NumPy (no deep learning framework). Open the notebook 'simple_neural_network_01.ipynb'. The tasks are:

- Run all cells in the notebook and make sure the neural network trains without errors.
- Explain in your own words (in a markdown cell) what the feature matrix (6x3) and the target (6x1) represent, and describe the pattern the network needs to learn.

- Report the actual predicted values for 'example_01' and 'example_02', and compare them to the expected ('Correct') values.
- Interpret the error-over-epoch plot: describe what happens to the error during training and why.
- Change the number of training epochs (e.g. from 2500 to 100) and describe, based on your own re-run, how the predictions and the error curve change.

Task 2

In this exercise, you will use a neural network implemented step by step to classify whether a person is diabetic, based on smoking, obesity, and exercise habits. Open the notebook 'simple_neural_network_02.ipynb'. The tasks are:

- Explain in your own words what each of the three features (Smoking, Obesity, Exercise) and the target (Diabetic) represent in the toy dataset.
- Run the step-by-step calculation cell and report the actual values you obtain for 'XW', 'sigmoid(XW) = z', the 'error', and the updated weights after this first manual update.
- Run the full training loop (200 epochs) and report the actual predicted probability of the new person (Smoking=0, Obesity=1, Exercise=0) being diabetic.
- Change the 'learning_rate' (e.g. to 0.05 and to 0.5) and describe, based on your own re-runs, how this affects the shape of the error-per-epoch curve and the final prediction.

Task 3

In this exercise, you will build and train a Convolutional Neural Network (CNN) with Keras/TensorFlow to recognize handwritten digits from the MNIST dataset. Open the notebook 'convolutional_neural_network_MNIST.ipynb'. The tasks are:

- Run all cells and report the actual shape of 'X_train' and 'X_test', and explain in your own words what each dimension represents.
- Change the index used to display a handwritten digit (e.g. from 2 to a number of your choice) and report which digit is shown.
- Describe the architecture of the CNN in your own words (number and type of layers, e.g. convolution, pooling, dropout, dense).
- Report the actual training accuracy and validation accuracy from the last training epoch, and the actual test accuracy and test loss you obtain when evaluating the model.

Task 4

In this exercise, you will build and train a neural network with Keras/TensorFlow to classify images of clothing from the Fashion MNIST dataset. Open the notebook 'convolutional_neural_network_MNIST_clothes.ipynb'. The tasks are:

- Run all cells and name at least 3 of the 10 clothing classes the model can distinguish.
- Report the actual training accuracy after the last (10th) epoch and the actual test accuracy, and explain in your own words what the gap between them indicates.
- Choose a different test image (change the index 'i' in the prediction cells) and report whether the model predicted correctly, and with what confidence (%).
- Explain in your own words the purpose of the 'Flatten' layer and the final 'Dense' layer with 10 outputs.